Following diagnosis of ectopic pregnancy in a married 25-year-old woman, an abdominal operation is performed. However, examination of the abdomen shows a normal uterine pregnancy but an acutely inflamed appendix. An appendectomy is performed despite the fact that the operative permit authorized only a surgical procedure for ectopic pregnancy. Which of the following best describes the surgeon's conduct?

- A) Appropriate in view of the medical necessity to remove the inflamed appendix
- B) Appropriate only if consent was obtained from the spouse
- C) Inappropriate, because it constituted an unauthorized extension of the field of surgery
- D) Inappropriate, because the patient had not previously been informed of the risks associated with an appendectomy

A previously healthy 37-year-old woman comes to the physician because of a 3-month history of headaches. The headaches now occur almost daily and are relieved by aspirin. Her pulse is 76/min, respirations are 12/min, and blood pressure is 158/95 mm Hg. No murmurs, gallops, or bruits are heard. Serum studies show:

Na^+	140 mEq/L
Cl^-	100 mEq/L
K^+	2.6 mEq/L
HCO_3^-	34 mEq/L
Urea nitrogen	9 mg/dL

Which of the following is the most likely diagnosis?

- A) Aldosteronoma
- B) Coarctation of the aorta
- C) Essential hypertension
- D) Pheochromocytoma
- E) Renal cell carcinoma

A 23-year-old woman has pain in the left shoulder 12 hours after falling down stairs. Examination shows diffuse abdominal tenderness. Hemoglobin concentration is 10 g/dL, and leukocyte count is 11,000/mm^3. Which of the following is the most likely diagnosis?

- A) Acute traumatic pancreatitis
- B) Diaphragmatic tear
- C) Ruptured spleen
- D) Shoulder separation
- E) Supracondylar fracture

A 67-year-old woman with end-stage renal disease, atherosclerotic coronary artery disease, and type 2 diabetes mellitus undergoes operative formation of an arteriovenous conduit in the left forearm for vascular access. The procedure is performed under an axillary block anesthesia. Twenty-four hours later, she has shortness of breath. Her temperature is 37.2°C (99°F), pulse is 129/min, respirations are 38/min and shallow, and blood pressure is 100/55 mm Hg. Examination shows jugular venous distention. Crackles are heard at both lung bases. S_1 and S_2 are normal. There is an S_3 and S_4. There is no edema of the sacrum or extremities. Administration of 300 mL of fluids is begun. Which of the following is the most likely diagnosis?

- A) Atelectasis
- B) High-output congestive heart failure
- C) Hypovolemia
- D) Pneumonia
- E) Pulmonary embolism

A 62-year-old man comes to the physician because he has been coughing up blood during the past week. He has a 5-year history of a daily productive cough. He has a 20-year history of hypertension well controlled with metoprolol. He has smoked two packs of cigarettes daily for 44 years. His blood pressure is 136/84 mm Hg. He is in no distress. Examination shows a barrel-shaped chest. His serum calcium concentration is 11 mg/dL. A chest x-ray shows a 4-cm lesion on the right lung. A biopsy specimen is most likely to show which of the following?

- A) Adenocarcinoma
- B) Mesothelioma
- C) Small cell lung carcinoma
- D) Squamous cell carcinoma
- E) Undifferentiated large cell carcinoma

Routine mammography on a 52-year-old woman shows six stippled microcalcifications in a cluster in the upper outer quadrant of the left breast; this finding was not present 1 year ago. No lump is palpable on examination of the breasts. She has no family history of breast cancer. Which of the following is the most appropriate next step in management?

- A) Follow-up mammography in 4 to 6 months
- B) Left upper outer quadrantectomy
- C) Fine-needle aspiration of the lesion
- D) Needle-localized open biopsy
- E) Thermography

Five days after undergoing an emergency colon resection for an obstructing carcinoma in the rectosigmoid, an 87-year-old man has the sudden onset of shortness of breath. He has a 15-year history of coronary artery disease. He is agitated and slightly confused. His temperature is 37.8°C (100°F), pulse is 104/min, respirations are 32/min and labored, and blood pressure is 88/50 mm Hg. Examination shows jugular venous distention. Crackles are heard over the midlung fields. Cardiac examination shows no murmurs or rubs. The abdomen is slightly distended. Bowel sounds are decreased. The liver span is 13 cm. Pulse oximetry on room air shows an oxygen saturation of 92%. An ECG shows ST-segment elevation in the anterior leads. Which of the following is the most likely diagnosis?

- A) Acute respiratory distress syndrome
- B) Cardiogenic shock
- C) Hypovolemic shock
- D) Pulmonary embolism
- E) Septic shock

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A 30-year-old man has had painless, progressive enlargement of the left testicle over the past 8 months. There is no nodule, and the scrotum cannot be transilluminated. Which of the following is the most likely diagnosis?

- A) Epididymitis
- B) Hydrocele
- C) Indirect inguinal hernia
- D) Spermatocele
- E) Testicular tumor

A 68-year-old man is brought to the emergency department because of recurrent vomiting of bright red blood and near syncope for 3 hours. He is afebrile. His pulse is 110/min, respirations are 16/min, and blood pressure is 90/60 mm Hg. He has mild epigastric tenderness. Which of the following is the most appropriate next step?

- A) Application of military antishock trousers
- B) Intravenous infusion of ranitidine
- C) Rapid infusion of 0.9% saline
- D) Gastric lavage with iced saline
- E) Emergency endoscopy

A 27-year-old woman who is HIV positive comes to the emergency department 4 hours after the onset of bloody diarrhea. She has had intermittent nonbloody diarrhea over the past 6 months. She has a 4-day history of abdominal cramps. She has had decreased appetite resulting in a 6.8-kg (15-lb) weight loss during the past week. She has had two hospitalizations for *Pneumocystis jiroveci* (formerly *P. carinii*) pneumonia during the past year. Current medications include didanosine, indinavir, stavudine, and trimethoprim-sulfamethoxazole. Her temperature is 39.4°C (103°F), pulse is 130/min, respirations are 24/min, and blood pressure is 80/60 mm Hg. Diffuse crackles are heard bilaterally. Examination shows a rigid, tympanitic, distended abdomen with diffuse rebound tenderness. Bowel sounds are decreased. Rectal examination shows gross blood. Her hemoglobin concentration is 8 g/dL, hematocrit is 24%, and leukocyte count is 3500/mm^3. After fluid resuscitation with 0.9% saline and transfusion of 2 units of packed red blood cells, she undergoes total abdominal colectomy with ileostomy for a perforated cecum. Pathologic examination of the excised colon shows diffuse mucosal inflammation with nuclear inclusion bodies. Which of the following is the most likely causal organism?

- A) *Campylobacter jejuni*
- B) *Cryptosporidium parvum*
- C) Cytomegalovirus
- D) *Entamoeba histolytica*
- E) *Mycobacterium avium-intracellulare*

A 46-year-old man with chronic alcoholism comes to the emergency department because of a 12-hour history of nausea and vomiting and midabdominal pain that radiates to the back. His pulse is 120/min, respirations are 20/min, and blood pressure is 110/60 mm Hg. Abdominal examination shows tenderness to palpation over the upper quadrants. Bowel sounds are absent. Laboratory studies show:

Hemoglobin	10.1 g/dL
Leukocyte count	24,500/mm^3
Platelet count	310,000/mm^3
Serum	
Amylase	1842 U/L
Albumin	4.1 g/dL
Calcium	7.7 mg/dL
Blood glucose	248 mg/dL

The most appropriate next step is intravenous administration of which of the following?

- A) Atropine
- B) Calcium gluconate
- C) Cimetidine
- D) Imipenem
- E) Lactated Ringer solution

A 77-year-old man comes to the emergency department 30 minutes after the onset of a severe nosebleed. He has collected 500 mL of blood and clots. He has a history of atherosclerotic cardiac disease, hypertension, and peripheral vascular disease. His medications include lisinopril and daily aspirin. His pulse is 110/min, respirations are 18/min, and blood pressure is 110/80 mm Hg. Anterior rhinoscopy shows no obvious source of the bleeding. Anterior nasal packing does not stop the bleeding; he now spits blood and clots out of his mouth. Which of the following vessels is the most likely source of this patient's bleeding?

- A) Ascending palatine artery
- B) Cavernous sinus
- C) Infraorbital artery
- D) Internal carotid artery
- E) Internal jugular vein
- F) Middle meningeal artery
- G) Sphenopalatine artery

A 24-year-old man is brought to the emergency department 30 minutes after he sustained burns to 50% of his total body surface area from boiling water. He is 178 cm (5 ft 10 in) tall and weighs 75 kg (165 lb); BMI is 24 kg/m^2. His pulse is 115/min, and blood pressure is 90/70 mm Hg. Examination shows blisters over the head, upper extremities, chest, and abdomen. There is erythema of the lower extremities. Which of the following is the most appropriate fluid resuscitation for this patient during the next 24 hours?

- A) 2 L of 5% albumin
- B) 6 L of 5% albumin
- C) 9.5 L of lactated Ringer solution
- D) 15 L of lactated Ringer solution
- E) 5 L of 3% saline
- F) 8 L of 3% saline

A hospitalized 72-year-old man has not voided since his urethral catheter was removed 8 hours ago. He underwent an elective sigmoid colectomy for recurrent diverticulitis 2 days ago. The colon was inflamed, and there was a hemorrhage during the dissection. He received 1 unit of packed red blood cells. His blood pressure did not decrease significantly during the operation; both ureters were visualized. He is currently receiving 5% dextrose in 0.45% saline (120 mL/h). His temperature is 38.2°C (100.8°F), pulse is 95/min, respirations are 14/min, and blood pressure is 145/80 mm Hg. Examination shows warm, dry skin. Abdominal examination shows mild distention of the lower quadrants; there is the suggestion of a mass. Which of the following is the most likely cause of this patient's inability to void?

- A) Acute tubular necrosis
- B) Bladder outflow obstruction
- C) Pigment-induced nephropathy
- D) Prerenal vasoconstriction
- E) Ureteral injury

Twelve hours after rod stabilization of a femoral shaft fracture, a 27-year-old homeless man has the sudden onset of combativeness and disorientation. His pulse is 120/min, respirations are 24/min, and blood pressure is 140/85 mm Hg. Examination shows petechiae over the axilla. The remainder of the examination shows no abnormalities. Which of the following is the most likely cause of these findings?

- A) Alcohol withdrawal
- B) Aspiration pneumonia
- C) Fat embolism
- D) Intracranial hemorrhage
- E) Pulmonary embolism
- F) Splenic rupture

Five days after emergency sigmoid colectomy for a perforated diverticulum, a 57-year-old man has a temperature of 40°C (104°F). Since the operation, he has had temperatures to 38.5°C (101.3°F). He has chronic obstructive pulmonary disease treated with corticosteroids and bronchodilators and has required repeated administration of inhaled bronchodilators to decrease his wheezing. He has been receiving cephalosporin since the operation. He was receiving 4 L/min of oxygen via nasal cannula to maintain an arterial oxygen saturation above 90%; he was recently switched to 50% oxygen by face mask to maintain an adequate saturation. Tubular breath sounds are heard at the left base. An x-ray of the chest shows a new infiltrate in the left lower lung field. A Gram stain of sputum shows many leukocytes and predominantly gram-negative rods. Which of the following is the most likely causal organism?

- A) *Candida albicans*
- B) Coagulase-negative *Staphylococcus* species
- C) *Enterococcus* species
- D) *Pseudomonas* species
- E) *Staphylococcus aureus*
- F) *Streptococcus pneumoniae*

A 47-year-old man comes to the physician because of a 2-day history of fever and increasingly severe rectal pain. He has type 2 diabetes mellitus treated with metformin. His temperature is 39.1°C (102.4°F), pulse is 90/min, respirations are 14/min, and blood pressure is 130/80 mm Hg. Examination shows a tender mass at the anal verge on the left. His hemoglobin concentration is 9.6 g/dL, leukocyte count is 18,000/mm^3, and serum glucose concentration is 350 mg/dL. Which of the following is the most appropriate next step in management?

- A) Sitz bath
- B) CT scan of the abdomen and pelvis
- C) Oral metronidazole therapy
- D) Flexible sigmoidoscopy
- E) Incision and drainage of the lesion

A 22-year-old man is brought to the emergency department after he sustained a gunshot wound to his right calf. On arrival, he is writhing in pain. His temperature is 37°C (98.6°F), pulse is 96/min, respirations are 16/min, and blood pressure is 110/76 mm Hg. Examination shows a 1-cm entrance wound over the medial aspect of the posterior right calf with no exit wound. The calf is tense and tender to palpation. Passive movement of the great toe exacerbates the patient's calf pain. Peripheral pulses are normal. Capillary refill time is 2 seconds. Sensation is intact. X-rays of the right lower extremity show a 1-cm bullet in the medial aspect of the posterior calf. Which of the following is the most appropriate immediate step in management?

- A) MRI of the right lower extremity
- B) Administration of antibiotics
- C) Irrigation of the wound
- D) Femoral arteriography
- E) Surgical decompression

A 52-year-old woman comes to the physician because of a 2-month history of swelling of her right foot. She has not had fever or pain. There is no history of trauma. She has a 25-year history of type 1 diabetes mellitus. Her temperature is 37°C (98.6°F). Examination shows swelling of the right foot over the dorsal forefoot and medial and lateral ankle regions with no ulcerations. There is bony hypertrophy and a small effusion over the ankle joint. Examination of the left foot shows no abnormalities. There is no redness, tenderness, or swelling in the calves or thighs. There is decreased sensation to light touch, proprioception, and pain below the ankles bilaterally. Peripheral pulses are present. An x-ray of the right ankle shows osteopenia and disruption and disorganization of the midtarsal and tarsometatarsal joints. Which of the following is the most likely cause of these findings?

- A) Arterial insufficiency of the lower extremity
- B) Bacterial infection with destruction of the joint spaces
- C) Deposition of monosodium urate crystals
- D) Lack of normal joint sensation
- E) Vasculitis

The driver of an automobile is brought to the emergency department following a head-on collision at 50 mph. He was wearing a seat belt. His only symptom is mild tenderness over the chest. All diagnostic tests show normal findings, except an x-ray of the chest that shows a somewhat widened mediastinum. Which of the following is the most appropriate next step in management?

- A) Aortic arteriography
- B) Bronchoscopy
- C) Discharge with a follow-up appointment in 24 hours
- D) Exploratory thoracotomy
- E) Observation in the hospital and another x-ray of the chest in 6 hours

A 42-year-old patient is admitted to the hospital with a piece of meat lodged in the lower esophagus. With considerable difficulty, the meat is removed by esophagoscopy. That evening, the patient has a temperature of 38.3°C (101°F). Which of the following is the most appropriate diagnostic study?

- A) CT scan of the chest
- B) Lateral x-rays of the neck
- C) MRI of the chest
- D) Repeated esophagoscopy
- E) Water-soluble contrast upper gastrointestinal study

Before undergoing elective cholecystectomy for acute cholecystitis, a 68-year-old man with chronic renal failure has the following laboratory findings:

Hematocrit	23%
Serum	
Na^+	130 mEq/L
Cl^-	94 mEq/L
K^+	5.4 mEq/L
HCO_3^-	12 mEq/L
Ca^{2+}	9.0 mg/dL
Urea nitrogen	90 mg/dL
Creatinine	7.5 mg/dL
Phosphorus	8.5 mg/dL

Arterial blood gas analysis shows a pH of 7.3. Which of the following is the most appropriate preoperative management?

- A) Low-protein diet
- B) Sodium polystyrene sulfonate (Kayexalate) enemas
- C) Administration of erythropoietin
- D) Administration of 1 L of 0.9% saline
- E) Initiation of dialysis

A 47-year-old man comes to the physician for evaluation prior to undergoing an abdominal perineal resection for rectal cancer. Over the past 6 years, he has had left ankle swelling. Examination shows mild edema of the left ankle without prominent veins. Venous duplex ultrasonography shows chronic occlusion of the left iliac and femoral veins. Immediately prior to the procedure, which of the following is the most appropriate management of this patient's venous disease?

- A) Aspirin prophylaxis
- B) Low-dose heparin prophylaxis
- C) Therapeutic heparin therapy
- D) Warfarin therapy
- E) Thrombolysis

A 6-year-old boy is brought to the emergency department 30 minutes after being involved in a motor vehicle collision. On arrival, he is awake and alert. There is an obvious scalp laceration over the left frontoparietal area. There are no neurologic deficits. A CT scan of the head shows a linear temporal skull fracture. Which of the following findings on CT scan is most likely in this patient?

- A) Epidural hematoma
- B) Subdural hematoma
- C) Intraventricular hemorrhage
- D) Subarachnoid hemorrhage
- E) Meningitis

The response options for the next **2** items are the same. Select one answer for each item in the set.

For each patient with a hernia, select the most likely diagnosis.

- A) Direct inguinal hernia
- B) Femoral hernia
- C) Incisional hernia
- D) Indirect inguinal hernia
- E) Obturator hernia
- F) Paraesophageal hiatal hernia
- G) Sliding hiatal hernia

A previously healthy 37-year-old woman comes to the physician because of left groin and thigh pain for 3 days. Examination shows a mass in the groin which is not reducible. Operative exploration shows a hernia sac medial to the femoral vein.

For each patient with a hernia, select the most likely diagnosis.

- A) Direct inguinal hernia
- B) Femoral hernia
- C) Incisional hernia
- D) Indirect inguinal hernia
- E) Obturator hernia
- F) Paraesophageal hiatal hernia
- G) Sliding hiatal hernia

A 62-year-old woman comes to the physician because of difficulty swallowing liquids and solid foods over the past month. She has a 10-year history of dyspepsia, substernal burning, and occasional nausea and vomiting. Examination shows no abnormalities. Esophagogastroscopy shows a stricture of the gastroesophageal junction and Barrett's esophagus.

An asymptomatic 72-year-old woman is admitted to the hospital because of decreased renal function. Her serum creatinine concentration has increased from 1.2 mg/dL to 3.3 mg/dL over the past 3 days. One month ago, she received a cadaveric renal transplant from a 65-year-old donor. This was her second transplant; the first was performed 25 years ago. Renal function had been excellent with the first allograft until 3 years ago when she was diagnosed with chronic rejection; she has been receiving hemodialysis since that time. Her original polycystic kidneys were removed prior to the first transplant, but the first allograft was not removed at the time of the second transplant. Current medications include prednisone, cyclosporine, and mycophenolate mofetil. Her temperature is 37°C (98.6°F), pulse is 76/min, respirations are 12/min, and blood pressure is 165/95 mm Hg. Examination shows no abnormalities. Which of the following is the most likely explanation for the decreased renal function in this patient?

- A) Age-related changes in the donor kidney
- B) Failure to remove the first renal allograft
- C) Failure to suppress class II antigen recognition in the host
- D) Failure to suppress host-circulating preformed reactive antibodies
- E) Recurrence of polycystic kidney disease in the organ transplant

A 42-year-old woman comes to the physician for a follow-up examination. She has a 3-month history of hypertension treated with a diuretic and a β-adrenergic blocking agent. At an examination 1 month ago, she had a blood pressure of 175/90 mm Hg. Her current pulse is 90/min, and blood pressure is 195/110 mm Hg. Abdominal examination shows a bruit heard best over the right midabdomen. Pelvic ultrasonography shows that the right kidney is smaller than the left. Arteriography shows fibromuscular hyperplasia of the right renal artery. Which of the following is the most likely mechanism of this patient's hypertension?

- A) Decreased extracellular fluid volume
- B) Decreased urine sodium retention
- C) Increased renin activity in both renal veins
- D) Increased renin activity in the left renal vein
- E) Increased serum aldosterone concentration

A 1-year-old boy is brought to the physician because of a persistent cough, loose stools, and a facial rash for 2 months. He has had a 1.8-kg (4-lb) weight loss during this period and is currently at the 50th percentile for length and 20th percentile for weight. His appetite and activity have remained normal, and there is no history of fever. His temperature is 37°C (98.6°F), pulse is 100/min, respirations are 18/min, and blood pressure is 80/50 mm Hg. Examination shows a malar flush and abdominal distention. CT scan of the abdomen shows a small tumor localized to the pancreas. The tumor is surgically removed, and the patient recovers uneventfully. Which of the following is the most likely diagnosis?

- A) Insulinoma
- B) Glucagonoma
- C) Neuroblastoma
- D) Pancreatic pseudocyst
- E) Vasoactive-inhibitory peptide-secreting tumor

Twelve hours after undergoing drainage of a pancreatic abscess, a 52-year-old man with alcoholism becomes bradycardic and hypoxic and requires intubation and mechanical ventilation. He weighs 70 kg (154 lb). The ventilator is set at an Fio_2 of 100%, tidal volume of 1000 mL, and positive end-expiratory pressure of 2.5 cm H_2O. Arterial blood gas analysis shows:

pH	7.36
Pco_2	40 mm Hg
Po_2	48 mm Hg

Which of the following is the most appropriate next step in management?

- A) Begin intravenous acetazolamide therapy
- B) Begin intravenous furosemide therapy
- C) Begin intravenous heparin therapy
- D) Begin intravenous sodium bicarbonate therapy
- E) Begin intravenous urokinase therapy
- F) Decrease Fio_2
- G) Decrease tidal volume
- H) Increase positive end-expiratory pressure
- I) Increase tidal volume

An 18-year-old woman comes to the physician for advice about screening tests because of concern about her family history of colon cancer. She has no history of serious illness and takes no medications. She says she feels well and has not had any gastrointestinal symptoms. Her father received the diagnosis of carcinoma of the sigmoid colon at the age of 40 years; genetic testing in her father showed the *APC* gene, which confirmed familial adenomatous polyposis. Examination of the patient today, including rectal examination, shows no abnormalities. Genetic testing shows the *APC* gene. On sigmoidoscopy, two 1-cm adenomatous polyps are removed. Which of the following is the most appropriate long-term management of this patient?

- A) Test of the stool for occult blood every 6 months
- B) Daily sulindac prophylaxis
- C) Colonoscopy every 5 years until the age of 40 years and every year after the age of 40 years
- D) Proctocolectomy with an ileoanal reservoir
- E) Right hemicolectomy
- F) No further management is indicated

A 60-year-old woman has a sigmoid resection and end colostomy for diverticulitis with rupture. That night she becomes confused, oliguric, and febrile. The tissues around the colostomy are indurated and crepitant. Which of the following is the most likely causal organism?

- A) *Bacteroides fragilis*
- B) *Clostridium perfringens*
- C) *Escherichia coli*
- D) Group A β-hemolytic streptococcus
- E) *Staphylococcus aureus*

. Thirty-six hours after admission to the hospital for evaluation of mild-to-moderate diffuse abdominal pain, a 42-year-old woman remains obstipated despite treatment with multiple laxatives. Her last bowel movement was 6 days ago. She has a 9-year history of systemic sclerosis (scleroderma) and has chronic constipation. She has a 1-year history of progressive abdominal distention. Her current temperature is 38°C (100.4°F), pulse is 110/min, respirations are 22/min, and blood pressure is 110/60 mm Hg. The abdomen is distended and tympanitic with diffuse tenderness to deep palpation. There is involuntary guarding and rebound. Her hemoglobin concentration is 12 g/dL, and leukocyte count is 14,000/mm^3. An abdominal x-ray shows dilation of the colon and a 12-cm cecum. There is no distention of the small bowel. Which of the following is the most appropriate next step in management?

- A) Metoclopramide therapy
- B) Octreotide therapy
- C) Colonoscopy
- D) Sigmoidoscopy
- E) Laparotomy

A 52-year-old woman is brought to the emergency department because of a 4-day history of progressive back pain and weakness and numbness in her legs. She has not urinated for the past 12 hours. She underwent a left mastectomy for breast cancer 5 years ago. Cranial nerves are intact. Muscle strength is full in the upper extremities and 3/5 to 4/5 diffusely in the lower extremities. Deep tendon reflexes are 2+ in the upper extremities and 3+ in the knees and ankles. Babinski sign is present bilaterally. Sensation to pinprick is decreased below the nipples, and sensation to vibration is absent in the lower extremities. She is unable to walk. Mental status examination shows no abnormalities. Insertion of a catheter yields 400 mL of urine. Which of the following is the most likely location of the abnormality?

- A) Cauda equina
- B) Cerebral hemispheres
- C) Cervical spinal cord
- D) Lumbar spinal cord
- E) Medulla
- F) Midbrain
- G) Pons
- H) Thalamus
- I) Thoracic spinal cord

Four days after undergoing coronary artery bypass grafting, a hospitalized 47-year-old man has the sudden onset of severe pain in his left great toe. He was admitted to the hospital 10 days ago for evaluation of chest pain with exertion, and received aspirin and heparin therapy for 3 days. Cardiac catheterization on admission showed coronary artery disease involving three vessels. The operation and initial postoperative course were uncomplicated. Vital signs are within normal limits. Examination shows a tender, cyanotic left great toe. There are new ecchymoses over the trunk and upper and lower extremities. The surgical incision appears clean and intact. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hematocrit	37%
Leukocyte count	12,200/mm^3
Platelet count	8000/mm^3
Prothrombin time	12 sec (INR=1)
Partial thromboplastin time	18 sec

Which of the following is the most likely diagnosis?

- A) Cholesterol embolization syndrome
- B) Disseminated intravascular coagulopathy
- C) Gout
- D) Heparin-induced thrombocytopenia
- E) Immune thrombocytopenic purpura

For 4 hours, a 55-year-old man has acute intermittent pain that begins in his right flank and radiates to his right testicle. His pulse is 90/min, and blood pressure is 145/95 mm Hg. Which of the following is the most likely finding on urinalysis?

- A) RBC casts
- B) Oval fat bodies
- C) pH 7.1
- D) Protein 1+
- E) RBC 30/hpf (in sediment)
- F) Specific gravity 1.002
- G) WBC 10/hpf

A 45-year-old man has had daily temperatures to 38.3°C (100.9°F) and a 7-kg (15-lb) weight loss over the past 3 months. He has pallor. His pulse is 84/min, respirations are 16/min, and blood pressure is 120/80 mm Hg. Lungs are clear to auscultation, and the precordium appears normal. Carotid upstroke is normal and equal bilaterally. S_1 is normal, and S_2 is physiologically split. A low-pitched, rumbling diastolic murmur is heard at the apex; the murmur disappears when the patient lies on his right side. The remainder of the examination is normal. Hemoglobin concentration is 10 g/dL. Which of the following is the most likely diagnosis?

- A) Aortic insufficiency
- B) Aortic stenosis
- C) Atrial myxoma
- D) Pericarditis
- E) Ventricular septal defect

A 47-year-old woman comes to the physician because of chafed skin on her inner thighs and armpits and under her breasts. Over the past 12 months, she has had panniculitis twice requiring admission to the hospital for intravenous administration of antibiotics. She is 173 cm (5 ft 8 in) tall and weighs 198 kg (438 lb); BMI is 67 kg/m^2. Examination of the inner thighs shows erythema and active fungal intertrigo. There is extensive scarring of the axillae and submammary areas. Pelvic examination shows a thick, white, curdy vaginal discharge. Examination of the perineum shows poor hygiene. Which of the following is the most appropriate long-term management for this patient?

- A) Oral fluconazole therapy
- B) Oral tetracycline therapy
- C) Sensitivity-guided antimicrobial therapy
- D) Panniculectomy
- E) Gastric bypass

A 52-year-old nulligravid woman comes to the physician because of a 2-month history of progressive abdominal swelling and decreased appetite. She has asthma treated with inhaled corticosteroids and type 2 diabetes mellitus treated with an oral hypoglycemic agent. Examination shows a markedly distended abdomen and a fluid wave. Bowel sounds are high pitched. A CT scan of the abdomen shows ascites, multiple pelvic masses, and omental thickening. Which of the following is the most likely diagnosis?

- A) Cervical cancer
- B) Choriocarcinoma
- C) Cirrhosis of the liver
- D) Endometrial carcinoma
- E) Leiomyomata uteri
- F) Ovarian adenocarcinoma

A previously healthy 47-year-old man comes to the physician because of a 2-week history of fatigue and progressive abdominal swelling. Abdominal examination shows distention and shifting dullness; bowel sounds are normal. There is no tenderness, masses, or organomegaly. A diagnostic paracentesis is performed and yields 50 mL of milky chylous fluid. Which of the following is the most likely cause of these findings?

- A) Budd-Chiari syndrome
- B) Chronic hepatitis B
- C) Lymphoma
- D) Mesothelioma
- E) Portal vein thrombosis

A previously healthy 37-year-old woman comes to the physician because of bruising on her arms and abdomen for 3 weeks. She takes ibuprofen as needed for occasional headaches. Her temperature is 36.9°C (98.4°F). Examination shows ecchymoses over the upper extremities and trunk. The lungs are clear to auscultation. Cardiac and abdominal examinations show no abnormalities. Laboratory studies show:

Hemoglobin	13.5 g/dL
Leukocyte count	6800/mm^3
Platelet count	45,000/mm^3
Bleeding time	11 min
Prothrombin time	15 sec (INR=1.4)

Examination of bone marrow aspirate shows increased megakaryocytes. Which of the following is the most likely explanation for these findings?

- A) Adverse effect of nonsteroidal anti-inflammatory drugs
- B) Bone marrow sequestration of platelets
- C) Circulating rheumatoid factor
- D) Formation of antiplatelet antibodies
- E) Previous cytomegalovirus infection

An 18-year-old man is brought to the emergency department because of pain in the right hip and flank after a motor vehicle collision. He is alert and stable. Examination shows abrasions, ecchymoses, and tenderness over the right iliac crest, and abrasions on the right flank. Urinalysis shows gross blood. An x-ray of the cervical spine shows normal findings. Which of the following is the most appropriate next step in diagnosis?

- A) CT scan of the abdomen with contrast
- B) MRI of the pelvis
- C) Radionuclide renal scan
- D) Selective renal angiography
- E) Cystoscopy

Twenty-four hours after removal of a solitary parathyroid adenoma for primary hyperparathyroidism, a 42-year-old woman has perioral numbness and tingling. Examination of the incision site shows no abnormalities. Her serum calcium concentration is 6.8 mg/dL, and serum albumin concentration is within normal limits. Which of the following is the most appropriate next step in management?

- A) Observation
- B) Intramuscular administration of magnesium
- C) Intravenous administration of calcitonin
- D) Intravenous administration of calcium gluconate
- E) Oral administration of vitamin D

A 13-year-old boy has had increasingly severe left thigh and knee pain and an associated limp for 2 days. There is no history of trauma. He is 183 cm (6 ft) tall and weighs 66 kg (145 lb); BMI is 20 kg/m^2. Passive motion of the hip elicits pain. An x-ray of the hips is shown. Which of the following is the most likely diagnosis?

- A) Avascular necrosis of the femoral head
- B) Juvenile rheumatoid arthritis
- C) Marfan syndrome
- D) Slipped capital femoral epiphysis
- E) Stress fracture

A 67-year-old postal worker comes to the physician 1 day after a 5-minute episode of weakness and numbness in his right (dominant) hand while at work. He did not have visual problems, headache, or weakness or numbness of the lower extremities. He is currently asymptomatic. He has smoked one pack of cigarettes daily for 45 years. He has not seen a physician in 40 years. His pulse is 85/min and irregular, and blood pressure is 140/90 mm Hg. The lungs are clear to auscultation. Pulses are palpable. Carotid bruits are heard bilaterally. Heart sounds are normal except for frequent premature beats. Neurologic examination shows no abnormalities. An ECG shows a normal sinus rhythm with multiple atrial premature contractions. Which of the following is the most appropriate next step in management?

- A) Echocardiography
- B) Carotid duplex ultrasonography
- C) Heparin therapy
- D) Tissue plasminogen activator therapy
- E) Warfarin therapy

A previously healthy 8-year-old boy is brought to the physician because of intermittent pain of his right anterior thigh over the past 3 weeks. His activity level is high and has remained unchanged during this period, but his mother notes that he limps intermittently. He is at the 25th percentile for height and weight. His temperature is 36.8°C (98.2°F), pulse is 76/min, respirations are 24/min, and blood pressure is 95/70 mm Hg. Examination shows an antalgic gait. Abduction and internal rotation of the femur is limited. X-rays of the hip are shown. Which of the following is the most likely mechanism of this child's condition?

- A) Avascular necrosis of the femoral head
- B) Bacterial infection of the bone
- C) Chronic inflammation of the synovium
- D) Congenital defect of the femoral head
- E) Malignant infiltration
- F) Mechanical slippage of the femoral head

A 62-year-old woman comes to the physician because of a 3-week history of progressive shortness of breath, mild pain in her right chest wall, and a nonproductive cough. She has had a 5.4-kg (12-lb) weight loss during the past 3 months. She received the diagnosis of carcinoma of the right breast 6 years ago and underwent lumpectomy followed by radiotherapy and chemotherapy. She has no other history of serious illness. She currently takes no medications. She is in mild respiratory distress and appears cyanotic and cachectic. She is 163 cm (5 ft 4 in) tall and weighs 52 kg (116 lb); BMI is 20 kg/m^2. Her pulse is 88/min, respirations are 24/min, and blood pressure is 114/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 90%. Examination shows tenderness to palpation over the right chest. Breath sounds are decreased on the right with dullness to percussion. A friction rub is heard at the right. Heart sounds are normal. Which of the following is the most likely diagnosis?

- A) Chest wall recurrence
- B) Empyema
- C) Pericardial effusion
- D) Pulmonary embolism
- E) Malignant pleural effusion

A 52-year-old woman with well-controlled type 2 diabetes mellitus is brought to the emergency department 24 hours after the onset of abdominal pain. She has vomited three times in the emergency department. She appears toxic. She is 157 cm (5 ft 2 in) tall and weighs 95 kg (210 lb); BMI is 38 kg/m^2. Her temperature is 39.5°C (103.1°F), pulse is 120/min, respirations are 28/min, and blood pressure is 160/90 mm Hg. Examination shows scleral icterus. Breath sounds are decreased over the lower lung fields bilaterally. The abdomen is mildly tender; bowel sounds are normal. Rectal examination shows no abnormalities. Laboratory studies show:

Hemoglobin	12.6 g/dL
Leukocyte count	16,000/mm^3
Segmented neutrophils	70%
Bands	10%
Lymphocytes	20%
Serum	
Na^+	130 mEq/L
Cl^-	90 mEq/L
K^+	5.0 mEq/L
HCO_3^-	14 mEq/L
Glucose	600 mg/dL
Bilirubin	
Total	5.2 mg/dL
Direct	2.0 mg/dL
Alkaline phosphatase	350 U/L
AST	300 U/L
Amylase	100 U/L

Which of the following is the most likely diagnosis?

- A) Acute pancreatitis
- B) Amebic liver abscess
- C) Biliary cirrhosis
- D) Cholangitis, ascending
- E) Chronic cholecystitis
- F) Hepatitis

A 67-year-old woman comes to the physician because of an 8-month history of increasing bilateral calf pain with walking. She initially had pain occasionally after a long walk; during the past 4 months, she has had pain after walking 300 feet on level ground or climbing one flight of stairs. The pain is relieved with rest. She has hypertension and hypercholesterolemia. Current medications include hydrochlorothiazide, metoprolol, and simvastatin. She smoked one pack of cigarettes daily for 30 years but stopped 1 year ago. Her pulse is 66/min, respirations are 14/min, and blood pressure is 132/81 mm Hg. Cardiopulmonary examination shows no abnormalities. Carotid, femoral, and dorsalis pedis pulses are decreased bilaterally. A soft femoral bruit is heard during systole bilaterally. There is no peripheral edema. Her serum electrolyte concentrations are within the reference range. Her serum urea nitrogen concentration is 28 mg/dL, and serum creatinine concentration is 2.1 mg/dL. Which of the following is the most appropriate next step in diagnosis?

- A) Determination of ankle brachial indices
- B) Duplex ultrasonography of the lower extremity veins
- C) Myelography of the lumbar spine
- D) CT scan of lumbosacral spine
- E) Arteriography of the lower extremities

An otherwise healthy 25-year-old woman comes to the physician because of a 2-month history of a pigmented lesion under the nail of her ring finger. A photograph of the lesion is shown. Which of the following is the most likely diagnosis?

- A) Actinic keratosis
- B) Angiosarcoma
- C) Basal cell carcinoma
- D) Compound nevus
- E) Congenital nevus
- F) Hemangioma
- G) Melanoma
- H) Subungual hematoma